Field Hours

CSCE 717 Algorithmic Game Theory

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Pure equilibrium and dominance

You describe a finite game by players, actions, and payoffs. In a 2-player game each cell holds two numbers, one payoff for the row player and one for the column player. A pure Nash equilibrium is a cell where neither player gains by switching alone. With cooperation paying 2 and mutual defection paying 1, defection is still the only equilibrium when a lone defector scores 3 and the cooperator scores 0.

You find that cell by dominance. If both cooperate, each scores 2. If you alone defect, you score 3 and the other scores 0. If both defect, each scores 1. Defection raises your score from 2 to 3 against cooperation and from 0 to 1 against defection. The unique pure equilibrium is mutual defection, payoff 1 each, even though both players prefer the pair where each scores 2.

Dominance picks the dilemma cell.

Both cooperate

Each scores 2

A loner can reach 3

Not an equilibrium

Both defect

Each scores 1

A switch scores 0

Pure equilibrium

Payoff when you alone defect

1. Both cooperate: each gets 2. You switch to defection: you get 3 and the other gets 0.

2. Both already defect: each gets 1. If you alone cooperate, your score falls from 1 to 0.

3. Defection is better against either action of the other player. The equilibrium pays 1 each, below the 2 each would earn by cooperating.

Problem 1. Both players cooperate and score 2. You defect and score 3. What is your gain, and what does it show?

Answer. The gain is 3 - 2 = 1. A one-step gain shows that mutual cooperation is not a pure Nash equilibrium.

Problem 2. Is mutual defection, payoff 1 each, a pure Nash equilibrium in this table?

Answer. Yes. A player who switches alone moves from 1 to 0, so neither gains by deviating.

Lecture 2 · 90 minutes

Mixed strategies

You allow a mixed strategy when no pure cell is stable. A mixed strategy assigns a probability to each of your actions. In matching pennies each player calls heads or tails. Matching pennies pays the row player 1 when the calls agree and -1 when they differ. The column player receives the opposite number. Every pure cell has a one-step gain for somebody, so no pure equilibrium exists. A distribution over calls is required.

You solve the mixture by making the opponent indifferent. Let the row player call heads with probability p. The column player then scores 1 - 2p by calling heads and 2p - 1 by calling tails. Set those payoffs equal and you obtain p = 1/2. The symmetric calculation forces the column player to use probability 1/2 as well. Each player expects payoff 0 at this equilibrium, and neither can rise above 0 by a unilateral change.

Indifference fixes the mixture.

1. Let row play heads at p

2. Equate the column payoffs

3. Solve and get p = 1/2

4. Expected payoff is 0

Matching pennies at 1/2

1. If row plays heads with probability p, column scores 1 - 2p by calling heads and 2p - 1 by calling tails.

2. Set 1 - 2p equal to 2p - 1. Then 2 = 4p, so p = 1/2.

3. Column also plays 1/2. Each player expects 0, and any pure call against 1/2 still expects 0.

Problem 1. Row plays heads with probability 1/2. What does column expect from calling heads, and from calling tails?

Answer. Heads scores 1 - 2*(1/2) = 0. Tails scores 2*(1/2) - 1 = 0.

Problem 2. Why is the cell where both players call heads not a pure equilibrium?

Answer. The calls match, so column receives -1 and can switch to tails to receive +1. That deviation gains 2.

Lecture 3 · 90 minutes

Price of anarchy on two roads

You measure inefficiency by the price of anarchy. Send one unit of traffic through Pigou's two-road network. One road has latency 1 no matter the load. The other has latency equal to its load x. Selfish drivers all choose the second road, since its latency stays below 1 until the load hits 1. At that equilibrium every driver has latency 1,

so total cost is 1. No driver gains by switching onto the constant road.

You compare that cost with an optimal split of the same unit. Put 1/2 of the traffic on each road. The constant road contributes cost 1/2. The load-dependent road has latency 1/2 and carries 1/2, contributing (1/2)*(1/2) = 1/4. Total optimal cost is 3/4. The price of anarchy is equilibrium cost over optimal cost, 1 divided by 3/4, which equals 4/3. This two-road network meets that ratio.

Price of anarchy on one unit of traffic.

One unit of traffic

Equilibrium cost is 1

Optimal cost is 3/4

Ratio is 4/3

Split the unit in half

1. At equilibrium all traffic uses the road of latency x. The load is 1 and the cost is 1.

2. Send 1/2 on the constant road, contribution 1/2, and 1/2 on the other road, contribution (1/2)*(1/2) = 1/4.

3. Optimal cost is 1/2 + 1/4 = 3/4. Divide 1 by 3/4 to get the price of anarchy 4/3.

Problem 1. Equilibrium cost is 1 and optimal cost is 3/4. What is the price of anarchy?

Answer. 1 / (3/4) = 4/3.

Problem 2. If you send the whole unit on the constant road, what is the total cost, and how does it compare with the optimum 3/4?

Answer. The latency is 1 on the whole unit, so the cost is 1. That is worse than 3/4 by 1 - 3/4 = 1/4.

Lecture 4 · 90 minutes

A voting cycle

You test a social choice rule on 3 candidates. A Condorcet winner would beat every other candidate by a head-to-head majority. Three voters can stop that from happening. Voter 1 ranks A over B over C. Voter 2 ranks B over C over A. Voter 3 ranks C over A over B. A majority prefers A to B, B to C, and C to A, each by 2 votes to 1. The pairwise tournament is a cycle of length 3.

You should not expect one rule to kill every cycle and still meet the usual axioms. Arrow's theorem says that once there are 3 or more candidates, a rule defined on every ranking profile, respecting unanimous pairwise wins, and ignoring irrelevant alternatives, must be a dictatorship. The 3-voter cycle is the obstruction you can write by hand. Exhibiting that cycle comes before any attempt to reconstruct the full proof.

Three majority votes, and no winner.

1. A beats B by 2 to 1

2. B beats C by 2 to 1

3. C beats A by 2 to 1

4. No Condorcet winner

Count the three majorities

1. A over B: voters 1 and 3 against voter 2, so 2 to 1.

2. B over C: voters 1 and 2 against voter 3, so 2 to 1.

3. C over A: voters 2 and 3 against voter 1, so 2 to 1. Each candidate loses one contest.

Problem 1. How many of these 3 voters prefer A to B, and how many prefer B to A?

Answer. Voters 1 and 3 prefer A to B, so 2. Voter 2 prefers B to A, so 1.

Problem 2. For 3 or more candidates, what does Arrow's theorem conclude about a rule that meets the axioms named in the lecture?

Answer. The rule must be a dictatorship. The 3-voter cycle is the pairwise obstruction you can check by hand.

Lecture 5 · 90 minutes

The second-price auction

You sell one indivisible item with a second-price auction. Each bidder submits one number. The highest bidder wins and pays the second-highest bid. With bids 9 and 5, the winner pays 5. If 9 is that winner's true value, utility is 9 - 5 = 4, while losing pays utility 0. The payment rule is the mechanism. It, rather than a request to be honest, makes the true value a dominant strategy.

You check the claim by testing a lie when your value is 9. Against a competing bid of 5, bidding 4 loses and your utility drops from 4 to 0. Bidding 12 still wins and still pays 5, so utility stays 4. Against a competing bid of 10, truthful bidding loses and scores 0. A bid of 12 wins, pays 10, and scores 9 - 10 = -1. The true bid avoids that negative utility. The other bidder may lie.

Second price charges the runner-up bid.

Collect the two bids -> Highest bid wins -> Charge the second bid -> Utility is value minus 5

Bids 9 and 5

1. The high bidder wins and pays 5, not 9.

2. If 9 is the true value, utility is 9 - 5 = 4. Losing would score 0.

3. Bidding 4 against the competing bid 5 loses, so utility falls from 4 to 0. Bidding the true 9 was better.

Problem 1. Your value is 9 and the other bid is 10. What is your utility if you bid 9, and if you bid 12?

Answer. Bidding 9 loses, utility 0. Bidding 12 wins and pays 10, utility 9 - 10 = -1. The truthful bid is better.

Problem 2. Bids are 9 and 5, and 9 is the true value. What does the winner pay, and what is the utility?

Answer. The winner pays the second price, 5. Utility is 9 - 5 = 4.

Lecture 6 · 90 minutes

A Vickrey-Clarke-Groves charge

You price bundles with the Vickrey-Clarke-Groves mechanism. Choose an allocation of maximum reported value, then charge each agent the harm imposed on the others. There are 2 items and 3 bidders. Bidder 1 values the pair at 15 and either single item at 0. Bidder 2 values item A at 6. Bidder 3 values item B at 6. Awarding both items to bidder 1 scores 15. Splitting the items scores 12, so the pair goes to bidder 1.

You read bidder 1's payment off that comparison. Without bidder 1 the best reported value is 12. Inside the chosen allocation the other bidders receive value 0. The Clarke charge is 12 - 0 = 12, and bidder 1's utility is 15 - 12 = 3. Bidders 2 and 3 get no item and pay 0. Removing either of them leaves the efficient award unchanged, so their charge stays 0. Truthful reporting of 15 is a dominant strategy.

Clarke payment for the bundle winner.

Pair value 15 beats 12

Others would score 12

Charge is 12

Utility is 3

Charge the bundle winner 12

1. Award both items to bidder 1 because 15 beats the split value 12.

2. Without bidder 1 the others score 12. In the chosen award they score 0. The payment is 12 - 0 = 12.

3. Utility is 15 - 12 = 3. Bidders 2 and 3 pay 0.

Problem 1. The chosen bundle scores 15 and the best allocation without bidder 1 scores 12. What are bidder 1's payment and utility?

Answer. Payment is 12 - 0 = 12. Utility is 15 - 12 = 3.

Problem 2. Why do bidders 2 and 3 pay 0 in this example?

Answer. Removing either one still awards both items to bidder 1, so each imposes harm 0 and the Clarke charge is 0.

Lecture 7 · 90 minutes

Existence versus search

You should not treat equilibrium existence as a cheap algorithm. Finding a mixed Nash equilibrium of a finite game is PPAD-complete. Solutions exist, because a continuous map from the mixed-strategy simplex to itself has a fixed point, and a fixed point is an equilibrium. Completeness means a polynomial algorithm for every such search would solve the rest of that class. The 2-player case stays PPAD-complete.

You can still finish a tiny game by listing supports. A support is the set of actions used with positive probability.

Two players with 3 actions each have 7 nonempty supports per player and therefore 49 support pairs. For one pair you solve a linear system: the opponent is indifferent inside the support and does not prefer to leave it. Matching pennies, with 2 actions, needs only a few systems. The step from 2 actions to 3 produces 49 pairs.

Existence and search are different claims.

Why one exists

A map on the simplex

Some fixed point

That point is equilibrium

Why listing is costly

2 players stay PPAD-complete

7 supports at 3 actions

49 support pairs

Count the support pairs

1. Each player has 3 actions, so 2^3 - 1 = 7 nonempty supports.

2. Two players give 7 * 7 = 49 support pairs.

3. Each pair is one linear system. Matching pennies has 2 actions per player and far fewer systems.

Problem 1. Two players have 3 actions each. How many nonempty supports does one player have, and how many pairs is that?

Answer. One player has 2^3 - 1 = 7 nonempty supports. The number of pairs is 7*7 = 49.

Problem 2. Does PPAD-completeness for 2-player Nash mean an equilibrium might fail to exist?

Answer. No. A fixed point of the mixed-strategy map still gives an equilibrium. Completeness is about the difficulty of the search.

Lecture 8 · 90 minutes

What you can now claim with a number

You can now start from a payoff table and finish with a number. You can name the dilemma equilibrium in which each player scores 1, and the matching-pennies equilibrium in which each player uses probability 1/2 and expects 0. You can compute the price of anarchy 4/3 for one unit of traffic on the two-road network. You can write the 3-voter cycle that blocks a Condorcet winner, and you can settle a one-item auction.

You can also separate a mechanism from a bare equilibrium. With bids 9 and 5, the second-price winner pays 5 and truthful bidding is dominant. In the two-item Vickrey-Clarke-Groves game the bundle winner pays 12 and keeps utility 3. You also know that 3 actions per player already give 49 support pairs, so a proof that an equilibrium exists is not the same object as a cheap procedure that returns one.

Numbers you can now defend.

Equilibrium payoffs 1 and 0

Road ratio 4/3

Payments 5 and 12

49 pairs are not an algorithm

A cycle beside a payment

1. The 3 voters make A beat B, B beat C, and C beat A, each by 2 to 1, so there is no Condorcet winner.

2. The second-price auction with bids 9 and 5 does name a winner, and that winner pays 5.

3. Pairwise majority can cycle. A payment rule can still select one bidder. You keep those conclusions separate.

Problem 1. Combine the two-road ratio with the auction payment. Costs 1 and 3/4 give price of anarchy 4/3, and bids 9 and 5 charge 5. Which number is a ratio of travel costs, and which is a payment?

Answer. 4/3 is 1 divided by 3/4, a ratio of travel costs. The charge 5 is the second bid. You do not fold the payment into the road ratio.

Problem 2. Combine the voting cycle with the bundle mechanism. The 3-voter profile has no Condorcet winner, but the bundle auction selects value 15 and charges 12. Why can the mechanism name a winner?

Answer. It maximizes a number, and 15 beats 12, then charges the harm 12. The majority tournament only stores pairwise votes of 2 to 1, and those votes cycle, so they name nobody.